

BUT R&T - IUT de Bethune

Compte rendu SAE3.03

Maquette reseau multi-sites (Siege + Succursale)

Informations document

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Document produit pour l'évaluation de la SAE3.03.

Mentions et perimetre

Objet

Ce rapport decrit la maquette reseau realisee dans le cadre de la SAE3.03. Il presente l'architecture, les choix techniques et les demonstrations attendues.

Perimetre

Deux sites (Siege et Succursale) avec segmentation par VLAN, administration controlee (SSH/ACL), services d'infrastructure (DHCP, DNS), proxy HTTP et centralisation des journaux.

Historique des versions

Version	Date	Evolution
2.0	06/02/2026	Refonte du rendu (lisibilite), extraits cibles, annexes allagees.

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1 Synthèse executive

Resume

La maquette met en oeuvre une segmentation par VLAN au siege et a la succursale, un controle d'accès pour l'administration (SSH/ACL), des mecanismes de redondance au siege (HSRP), ainsi que les services attendus (DHCP, DNS, proxy HTTP et centralisation des logs). Les annexes integrent des extraits de configuration et des preuves disponibles. Pour les points a demontrer en seance, une procedure de test reproductible est fournie (aucun resultat n'est invente).

Table de conformite (vue rapide)

Exigence	Element	Support
Reseau ADMIN	VLAN ADMIN, reseau management, acces aux equipements	Extraits + tests
SSH depuis ADMIN uniquement	ACL SSH_ADMIN, lignes VTY, test croise (ADMIN OK / autres VLAN KO)	Extraits + tests
Segmentation (PERSONNEL / PROD / WIFI / VIDEO)	VLAN et SVI, plan d'adressage, filtrage inter-VLAN selon besoin	Extraits + tests
Redondance siege	HSRP (VIP), verification des etats, test de bascule	Preuve + test
VPN siege-succursale	IPsec (SA, compteurs), verification et trafic inter-sites	Config + test
Succursale via siege / proxy	Traceroute et acces Web via proxy, trace dans access.log	Config + test
Services DHCP / DNS	Attribution DHCP et resolution DNS (NOERROR)	Preuves partielles + tests
Logs centralises	Emission syslog et reception sur serveur (rsyslog)	Preuve service + test

2 Contexte et objectifs

2.1 Contexte

L'entreprise est organisee autour d'un **siege** et d'une **succursale distante**. Les echanges inter-sites doivent etre securises et l'administration des equipements doit etre maitrisee.

2.2 Objectifs

Acces SSH uniquement depuis le reseau ADMIN, redondance au siege, segmentation (PERSONNEL/VIDEO/PRODUCTION/WIFI), VPN siege-succursale, trafic succursale achemine via le siege et proxy, services DNS/DHCP et centralisation des journaux.

3 Repartition des taches

A completer avant depot

Le dossier d'évaluation demande une repartition des taches par membre du binome. Compléter le tableau avec les realisations effectives (conception, configuration, tests, documentation).

Auteur Taches realisees

4 Architecture et topologie

4.1 Vue d'ensemble

Le siege s'appuie sur deux commutateurs L3 assurant le routage inter-VLAN et la redondance de passerelle (HSRP), ainsi que des commutateurs d'accès. La succursale est interconnectée via un routeur et un lien WAN.

4.2 Capture de la topologie

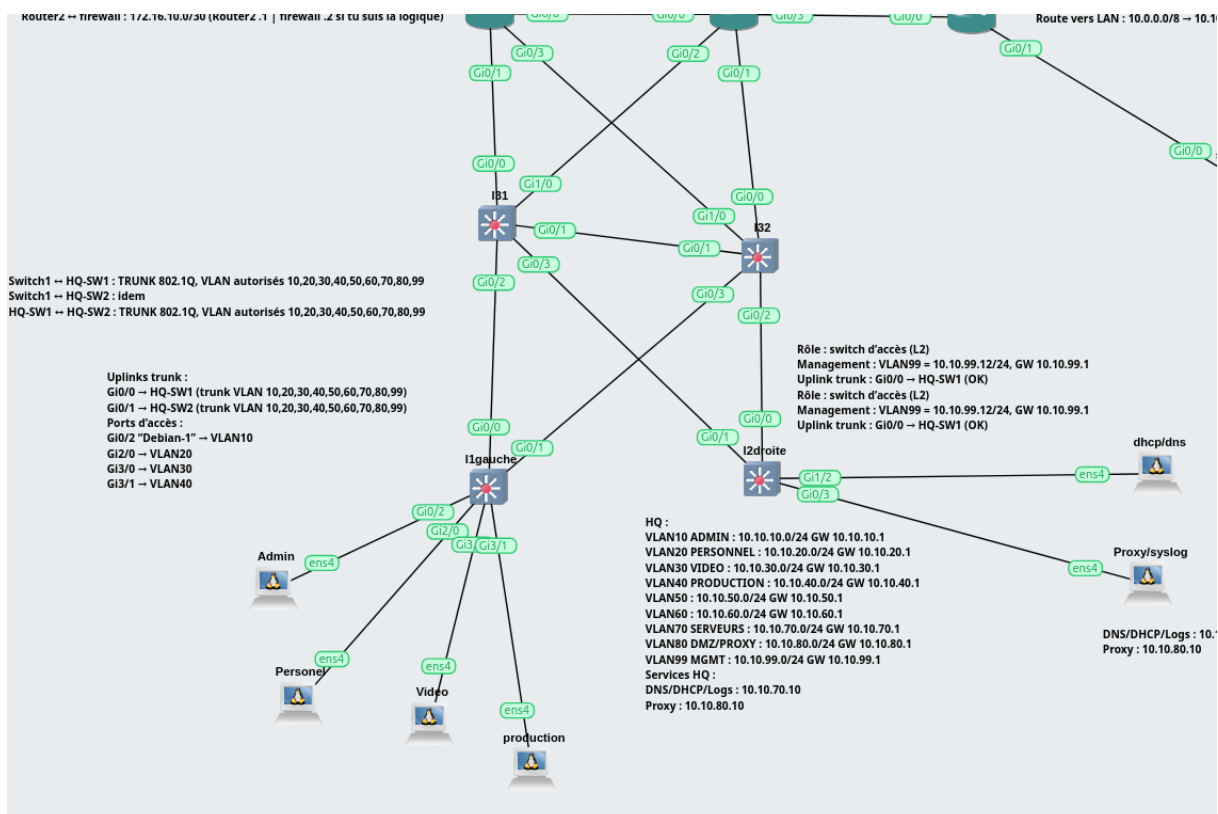


FIGURE 1 – Topologie de la maquette.

5 Plan d'adressage

6 Conventions

Les VLANs utilisateurs et serveurs du siege utilisent le plan 10.10.VLAN.0/24. Le VLAN de management est separe (10.0.99.0/24). La succursale utilise 10.20.VLAN.0/24.

7 Siege : VLANs L3 et passerelles HSRP

VLAN	Usage	Sous-reseau	VIP	HQ-SW1	HQ-SW2
10	ADMIN	10.10.10.0/24	10.10.10.254	10.10.10.1	10.10.10.2
20	PERSONNEL	10.10.20.0/24	10.10.20.254	10.10.20.1	10.10.20.2
30	VIDEO	10.10.30.0/24	10.10.30.254	10.10.30.1	10.10.30.2
40	PRODUCTION	10.10.40.0/24	10.10.40.254	10.10.40.1	10.10.40.2
50	WIFI (personnel)	10.10.50.0/24	10.10.50.254	10.10.50.1	10.10.50.2
60	WIFI (invites)	10.10.60.0/24	10.10.60.254	10.10.60.1	10.10.60.2
70	SERVEURS	10.10.70.0/24	10.10.70.254	10.10.70.1	10.10.70.2
80	DMZ / PROXY	10.10.80.0/24	10.10.80.254	10.10.80.1	10.10.80.2
99	MGMT	10.0.99.0/24	10.0.99.254	10.0.99.1	10.0.99.2

8 Succursale : VLANs (router-on-a-stick)

VLAN	Usage	Sous-reseau	Passerelle (BR-R1)
10	ADMIN	10.20.10.0/24	10.20.10.1
20	PERSONNEL	10.20.20.0/24	10.20.20.1
30	VIDEO	10.20.30.0/24	10.20.30.1
40	PRODUCTION	10.20.40.0/24	10.20.40.1
50	WIFI (personnel)	10.20.50.0/24	10.20.50.1
60	WIFI (invites)	10.20.60.0/24	10.20.60.1
99	MGMT	10.20.99.0/24	10.20.99.1

9 Liaisons WAN

Lien	Reseau	Adresses
Siege (Router1) ↔ Succursale (BR-R1)	10.10.250.16/30	Router1 : 10.10.250.17, BR-R1 : 10.10.250.18
Router2 ↔ Routeur "inter-net"	10.10.250.8/30	Router2 : 10.10.250.10, internet : 10.10.250.9
Router2 ↔ Firewall	172.16.10.0/30	Router2 : 172.16.10.1 (pair non fournie)

10 Segmentation et redondance au siege

10.1 Segmentation par VLAN

Segmentation par VLAN dedies aux usages (ADMIN, PERSONNEL, VIDEO, PRODUCTION, WIFI, SERVEURS, DMZ) et un VLAN management.

Preuve : VLAN declares

```

1 Q-SW2#show vlan brief
2
3 VLAN Name                Status    Ports
4 -----
5 1    default                active    Gi1/2, Gi1/3, Gi2/0, Gi2/1
6                                     Gi2/2, Gi2/3, Gi3/0, Gi3/1
7                                     Gi3/2, Gi3/3
8 10   VLAN10_ADMIN           active
9 20   VLAN20_PERSONNEL       active
10 30   VLAN30_VIDEO           active
11 40   VLAN40_PROD            active
12 50   VLAN50_WIFI_PERSO      active
13 60   VLAN60_WIFI_GUEST      active
14 70   VLAN70_SERVEURS        active
15 80   VLAN80_DMZ             active
16 99   VLAN99_MGMT            active    Gi0/0, Gi1/0
17 1002 fddi-default           act/unsup
18 1003 token-ring-default     act/unsup
19 1004 fddinet-default        act/unsup
20 1005 trnet-default

```

10.2 Interfaces L3 (SVI) et relais DHCP

Preuve : SVI en etat up/up

```

1 Vlan10          10.10.10.2    YES manual up        up
2 Vlan20          10.10.20.2    YES manual up        up
3 Vlan30          10.10.30.2    YES manual up        up
4 Vlan40          10.10.40.2    YES manual up        up
5 Vlan50          10.10.50.2    YES manual up        up
6 Vlan60          10.10.60.2    YES manual up        up
7 Vlan70          10.10.70.2    YES manual up        up
8 Vlan80          10.10.80.2    YES manual up        up
9 Vlan99          10.0.99.2      YES NVRAM  up        up
10
11
12 Vlan70          10.10.70.1    YES manual up        up
13 Vlan80          10.10.80.1    YES manual up        up
14 Vlan99          10.0.99.1      YES NVRAM  up        up
15
16
17 Vlan99          10.0.99.11    YES manual up        up

```

Extrait : relais DHCP (ip helper-address)

```

1 interface Vlan10
2 ip helper-address 10.10.70.10
3 !
4 interface Vlan20
5 !
6 interface Vlan70
7 ip helper-address 10.10.70.10

```



```

8 !
9 interface Vlan80
10 ip helper-address 10.10.70.10

```

10.3 Redondance de passerelle (HSRP)

Preuve : etat HSRP (VIP et roles)

```

1 HQ-SW2#show standby brief
2           P indicates configured to preempt.
3           |
4 Interface    Grp  Pri P State    Active        Standby        Virtual IP
5 V110         10   100 P Standby  10.10.10.1    local          10.10.10.254
6 V120         20   100 P Standby  10.10.20.1    local          10.10.20.254
7 V130         30   100 P Standby  10.10.30.1    local          10.10.30.254
8 V140         40   100 P Standby  10.10.40.1    local          10.10.40.254
9 V150         50   100 P Standby  10.10.50.1    local          10.10.50.254
10 V160         60   100 P Standby  10.10.60.1    local          10.10.60.254
11 V170         70   100 P Standby  10.10.70.1    local          10.10.70.254
12 V180         80   100 P Standby  10.10.80.1    local          10.10.80.254
13 V199         99   100 P Standby  10.0.99.1     local          10.0.99.254

```

10.4 STP

Extrait : parametres STP

```

1 spanning-tree mode rapid-pvst
2 spanning-tree extend system-id
3 spanning-tree vlan 10,20,30,40,50,60,70,80,99 priority 24576

```

11 Services (DNS, DHCP, proxy, logs)

11.1 DHCP

Configuration (serveur)

Configuration dans configs_services/.

Preuve : attribution DHCP (client)

```

1 debian@debian:~$ sudo dhclient -v ens4
2 ip a
3 ip route
4 ping gw.hq.local
5 ping google.com
6 Internet Systems Consortium DHCP Client 4.4.3-P1
7 Copyright 2004-2022 Internet Systems Consortium.
8 All rights reserved.
9 For info, please visit https://www.isc.org/software/dhcp/
10
11 Listening on LPF/ens4/0c:ad:6c:13:00:00
12 Sending on   LPF/ens4/0c:ad:6c:13:00:00
13 Sending on   Socket/fallback
14 DHCPDISCOVER on ens4 to 255.255.255.255 port 67 interval 5

```

```
15 DHCPPOFFER of 10.0.10.50 from 10.0.10.10
16 DHCPREQUEST for 10.0.10.50 on ens4 to 255.255.255.255 port 67
17 DHCPACK of 10.0.10.50 from 10.0.10.10
18 bound to 10.0.10.50 -- renewal in 245 seconds.
```

11.2 DNS (BIND)

Configuration (serveur)

Configuration BIND dans configs_services/.

Preuve : service named actif

```
1 named.service - BIND Domain Name Server
2   Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enable>
3   Active: active (running) since Wed 2026-02-04 17:35:44 UTC; 32s ago
4     Docs: man:named(8)
5   Main PID: 415 (named)
6     Status: "running"
7     Tasks: 5 (limit: 537)
8    Memory: 25.5M
9         CPU: 1.417s
10    CGroup: /system.slice/named.service
11           415 /usr/
```

Preuve : test de resolution (capture)

```
1 dig @127.0.0.1 dns.hq.local
2 dig @127.0.0.1 google.com
3
4 ; <<>> DiG 9.18.44-1-deb12u1-Debian <<>> @127.0.0.1 dns.hq.local
5 ; (1 server found)
6 ;; global options: +cmd
7 ;; Got answer:
8 ;; WARNING: .local is reserved for Multicast DNS
9 ;; You are currently testing what happens when an mDNS query is leaked to DNS
10 ;; ->>HEADER<<- opcode: QUERY, status: REFUSED, id: 34831
11 ;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
12 ;; WARNING: recursion requested but not available
13
14 ;; OPT PSEUDOSECTION:
```

11.3 Proxy HTTP (Squid)

Configuration (serveur)

Configuration Squid dans configs_services/squid.conf.

Preuve : autorisation d'accès au proxy (ACL runtime)

```
1 HQ-SW2#show access-lists LAN_TO_PROXY
2 Extended IP access list LAN_TO_PROXY
3   10 permit udp any 224.0.0.0 0.0.0.255 (46 matches)
4   20 permit icmp any any
5   30 permit udp host 10.10.10.1 host 10.10.70.10 eq bootps
```

```
6      40 permit udp host 10.10.10.2 host 10.10.70.10 eq bootps
7      50 permit udp host 10.10.70.10 host 10.10.10.1 eq bootps
8      60 permit udp host 10.10.70.10 host 10.10.10.2 eq bootps
9      70 permit udp any eq bootps any eq bootpc
10     80 permit tcp 10.10.0.0 0.0.255.255 host 10.10.80.10 eq 3128
11     90 deny ip any any log (1 match)
```

11.4 Centralisation des logs

Extrait : configuration Syslog sur équipements

```
1 logging source-interface Vlan99
2 logging host 10.0.10.10
3 service timestamps debug datetime msec
4 service timestamps log datetime msec
```

Preuve : service rsyslog actif

```
1 * rsyslog.service - System Logging Service
2   Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
3   Active: active (running) since Wed 2026-02-04 17:44:17 UTC; 504ms ago
4   TriggeredBy: * syslog.socket
5     Docs: man:rsyslogd(8)
6           man:rsyslog.conf(5)
7           https://www.rsyslog.com/doc/
8   Main PID: 528 (rsyslogd)
9     Tasks: 7 (limit: 537)
10    Memory: 908.0K
11       CPU: 133ms
12    CGroup: /system.slice/rsyslog.service
13           528 /usr/sbin/rsyslogd -n -iNONE
```

12 Contrôle d'accès et administration (SSH, ACL)

12.1 SSH restreint au réseau ADMIN

Extrait : configuration (ACL SSH_ADMIN + VTY)

```
1 ip access-list standard SSH_ADMIN
2   permit 10.10.10.0 0.0.0.255
3   deny   any
4   !
5 line vty 0 4
6   access-class SSH_ADMIN in
7   login local
8   transport input ssh
9 line vty 5 15
10  access-class SSH_ADMIN in
11  login local
12  transport input ssh
13  !
14 line vty 5 15
15  access-class SSH_ADMIN in
16  login local
17  transport input ssh
```

Preuve : ACL SSH_ADMIN en runtime

```

1 HQ-SW1#show access-lists SSH_ADMIN
2 Standard IP access list SSH_ADMIN
3     10 permit 10.0.10.0, wildcard bits 0.0.0.255
4     20 permit 10.0.99.0, wildcard bits 0.0.0.255
5     30 deny any (3 matches)
6     40 permit 10.10.10.0, wildcard bits 0.0.0.255
7 HQ-SW1#

```

Preuve : tentative bloquee (journal ACL)

```

1 *Feb  4 23:14:20.924: %SEC-6-IPACCESSLOGP: list LAN_TO_PROXY denied tcp 10.10.10.50(59932) ->
  10.0.99.1(22), 1 packet

```

12.2 DMZ et journalisation

Extrait : ACL DMZ

```

1 ip access-list extended DMZ_IN
2 permit tcp any any established
3 permit tcp host 10.10.80.10 any eq www
4 permit tcp host 10.10.80.10 any eq 443
5 permit udp host 10.10.80.10 any eq domain
6 permit tcp 10.10.10.0 0.0.0.255 host 10.10.80.10 eq 3128
7 permit tcp 10.10.30.0 0.0.0.255 host 10.10.80.10 eq 3128
8 permit tcp 10.10.20.0 0.0.0.255 host 10.10.80.10 eq 3128
9 permit udp host 10.10.80.10 host 10.10.70.10 eq domain
10 deny ip any any log
11 remark === DNS vers serveur interne ===
12 permit udp 10.10.80.0 0.0.0.255 host 10.10.70.10 eq domain
13 permit tcp 10.10.80.0 0.0.0.255 host 10.10.70.10 eq domain
14 remark === HTTP/HTTPS vers Internet ===
15 permit tcp 10.10.80.0 0.0.0.255 any eq www
16 permit tcp 10.10.80.0 0.0.0.255 any eq 443
17 remark === ICMP debug (optionnel) ===
18 permit udp host 10.10.80.10 any eq ntp
19 ip access-list extended LAN_TO_PROXY
20 remark === Multicast (HSRP, routing interne)
21 permit udp any 224.0.0.0 0.0.0.255
22 remark === ICMP (debug, supervision)
23 permit icmp any any
24 remark === DHCP (clients)
25 permit udp any eq bootpc any eq bootps
26 permit udp any eq bootps any eq bootpc
27 remark === DHCP relay vers serveur
28 permit udp host 10.10.10.1 host 10.10.70.10 eq bootps
29 permit udp host 10.10.10.2 host 10.10.70.10 eq bootps
30 permit udp host 10.10.70.10 host 10.10.10.1 eq bootps
31 permit udp host 10.10.70.10 host 10.10.10.2 eq bootps
32 remark === DNS vers serveur interne
33 permit udp 10.10.0.0 0.0.255.255 host 10.10.70.10 eq domain
34 permit tcp 10.10.0.0 0.0.255.255 host 10.10.70.10 eq domain
35 remark === SSH ADMIN (ALLER)
36 permit tcp 10.10.10.0 0.0.0.255 10.0.99.0 0.0.0.255 eq 22
37 remark === SSH ADMIN (RETOUR)
38 permit tcp 10.0.99.0 0.0.0.255 10.10.10.0 0.0.0.255 established
39 remark === Proxy Squid
40 permit tcp 10.10.0.0 0.0.255.255 host 10.10.80.10 eq 3128
41 remark === DENY FINAL
42 deny ip any any log

```

Preuve : logs ACL DMZ

```

1 *Feb  4 22:37:12.228: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(37866) ->
   176.137.36.37(123), 1 packet
2 *Feb  4 22:37:22.500: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(38739) ->
   5.39.80.51(123), 1 packet
3 *Feb  4 22:37:32.703: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(58751) ->
   141.95.158.40(123), 1 packet
4 *Feb  4 22:37:39.458: %SEC-6-IPACCESSLOGRL: access-list logging rate-limited or missed 4 packets
5 *Feb  4 22:37:42.431: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.1(1985) ->
   224.0.0.2(1985), 134 packets
6 *Feb  4 23:07:18.531: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(44903) ->
   82.66.40.79(123), 1 packet

```

13 Interconnexion siege-succursale et acheminement

13.1 Routage (WAN)

Extrait : routes

```

1 ip route 0.0.0.0 0.0.0.0 10.10.250.18
2 ip route 0.0.0.0 0.0.0.0 10.10.250.9
3 ip route vrf VRF-SITE 0.0.0.0 0.0.0.0 10.10.250.9
4 ip route 0.0.0.0 0.0.0.0 10.10.250.17
5 ip route 10.10.0.0 255.255.0.0 GigabitEthernet0/0

```

13.2 VPN IPsec

Extrait : configuration IPsec (objets principaux)

```

1 crypto isakmp policy 10
2   encr aes
3   authentication pre-share
4   group 14
5 crypto isakmp key VPN-SITE address 10.10.250.18
6 !
7 crypto isakmp key VPN-SITE address 10.10.250.18
8 !
9 crypto ipsec transform-set TS-SITE esp-aes esp-sha-hmac
10 mode tunnel
11 !
12 crypto map CMAP-SITE 10 ipsec-isakmp
13   set peer 10.10.250.18
14   set transform-set TS-SITE
15   match address VPN-SITE
16 !
17 ip access-list extended VPN-SITE
18   permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255

```

Preuve : etat IPsec (capture)

```

1 Router2#show crypto isakmp sa
2 IPv4 Crypto ISAKMP SA
3 dst          src          state          conn-id status
4
5 IPv6 Crypto ISAKMP SA
6

```

```
7 Router2#show crypto ipsec sa
8
9 interface: GigabitEthernet0/3
10   Crypto map tag: CMAP-SITE, local addr 10.10.250.10
11
12   protected vrf: VRF-SITE
13   local ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
14   remote ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
15   current_peer 10.10.250.18 port 500
16     PERMIT, flags={origin_is_acl,}
17     #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
18     #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
19     #pkts compressed: 0, #pkts decompressed: 0
20     #pkts not compressed: 0, #pkts compr. failed: 0
21     #pkts not decompressed: 0, #pkts decompress failed: 0
22     #send errors 0, #recv errors 0
23
24   local crypto endpt.: 10.10.250.10, remote crypto endpt.: 10.10.250.18
25
26 Router2#ping vrf VRF-SITE 10.20.0.1
27 Type escape sequence to abort.
28 Sending 5, 100-byte ICMP Echos to 10.20.0.1, timeout is 2 seconds:
29 .....
30 Success rate is 0 percent (0/5)
31 Router2#
```

13.3 Succursale : sous-interfaces VLAN

Extrait : sous-interfaces

```
1 interface GigabitEthernet0/1.10
2   encapsulation dot1Q 10
3   ip address 10.20.10.1 255.255.255.0
4   !
5 interface GigabitEthernet0/1.20
6   encapsulation dot1Q 20
7   ip address 10.20.20.1 255.255.255.0
8   !
9 interface GigabitEthernet0/1.30
10  encapsulation dot1Q 30
11  ip address 10.20.30.1 255.255.255.0
12  !
13 interface GigabitEthernet0/1.40
14  encapsulation dot1Q 40
15  ip address 10.20.40.1 255.255.255.0
16  !
17 interface GigabitEthernet0/1.50
18  encapsulation dot1Q 50
19  ip address 10.20.50.1 255.255.255.0
20  !
21 interface GigabitEthernet0/1.60
22  encapsulation dot1Q 60
23  ip address 10.20.60.1 255.255.255.0
24  !
25 interface GigabitEthernet0/1.99
26  encapsulation dot1Q 99
27  ip address 10.20.99.1 255.255.255.0
```

13.4 Succursale via siege + proxy

Demontrer un `traceroute` depuis la succursale vers une IP publique, puis un acces Web via proxy avec une entree dans `access.log`. Voir “Batteries de tests”.

14 Priorisation du trafic video (QoS)

15 Principe

L'exigence impose de privilegier le trafic VIDEO (VLAN30) sur l'ensemble du reseau.

16 Configuration / proposition

Les elements fournis disponibles contiennent des elements de configuration QoS (class-map/policy-map). Une mise en uvre complete necessite (selon plateforme) l'activation QoS et l'application de la policy sur les interfaces pertinentes.

Elements presents dans les elements fournis (extraits)

```
1 class-map match-any VIDEO_TRAFFIC match vlan 30 policy-map QOS_HQ class VIDEO_TRAFFIC set dscp
  af41 priority percent 30
```

17 Validation

La validation se fait par inspection de la policy (compteurs) et par generation de trafic concurrent (ex. iperf) afin d'observer la priorisation. La procedure est fournie en §10.

18 Batteries de tests (demonstrations)

Principe

Cette section decrit des demonstrations reproductibles pour chaque exigence du sujet. Les **preuves disponibles** sont integrees quand elles existent. Lorsqu'une preuve n'est pas presente dans les elements fournis, elle est indiquee **a produire en seance** (commande a executer + sortie a capturer). Aucun resultat n'est invente.

18.1 Reseau ADMIN

Procedure (poste ADMIN)

```
1 ip a
2 ip route
3 ping -c 3 10.10.10.254
4 ssh admin@10.0.99.1
```

Preuves disponibles

VLAN declares :

```
1 Q-SW2#show vlan brief
2
3 VLAN Name                               Status    Ports
4 -----                               -
```

```

5 1    default                                active    Gi1/2, Gi1/3, Gi2/0, Gi2/1
6                                           Gi2/2, Gi2/3, Gi3/0, Gi3/1
7                                           Gi3/2, Gi3/3
8 10   VLAN10_ADMIN                          active
9 20   VLAN20_PERSONNEL                      active
10 30   VLAN30_VIDEO                          active
11 40   VLAN40_PROD                           active
12 50   VLAN50_WIFI_PERSO                     active
13 60   VLAN60_WIFI_GUEST                     active
14 70   VLAN70_SERVEURS                       active
15 80   VLAN80_DMZ                             active
16 99   VLAN99_MGMT                           active    Gi0/0, Gi1/0
17 1002 fddi-default                          act/unsup
18 1003 token-ring-default                    act/unsup
19 1004 fddinet-default                       act/unsup
20 1005 trnet-default

```

SVI up/up :

```

1 Vlan10      10.10.10.2    YES manual up          up
2 Vlan20      10.10.20.2    YES manual up          up
3 Vlan30      10.10.30.2    YES manual up          up
4 Vlan40      10.10.40.2    YES manual up          up
5 Vlan50      10.10.50.2    YES manual up          up
6 Vlan60      10.10.60.2    YES manual up          up
7 Vlan70      10.10.70.2    YES manual up          up
8 Vlan80      10.10.80.2    YES manual up          up
9 Vlan99      10.0.99.2     YES NVRAM up            up
10
11
12 Vlan70      10.10.70.1    YES manual up          up
13 Vlan80      10.10.80.1    YES manual up          up
14 Vlan99      10.0.99.1     YES NVRAM up            up
15
16
17 Vlan99      10.0.99.11    YES manual up          up

```

HSRP (VIP et roles) :

```

1 HQ-SW2#show standby brief
2                               P indicates configured to preempt.
3                               |
4 Interface  Grp  Pri  P State  Active      Standby      Virtual IP
5 V110       10   100 P Standby 10.10.10.1   local        10.10.10.254
6 V120       20   100 P Standby 10.10.20.1   local        10.10.20.254
7 V130       30   100 P Standby 10.10.30.1   local        10.10.30.254
8 V140       40   100 P Standby 10.10.40.1   local        10.10.40.254
9 V150       50   100 P Standby 10.10.50.1   local        10.10.50.254
10 V160       60   100 P Standby 10.10.60.1   local        10.10.60.254
11 V170       70   100 P Standby 10.10.70.1   local        10.10.70.254
12 V180       80   100 P Standby 10.10.80.1   local        10.10.80.254
13 V199       99   100 P Standby 10.0.99.1    local        10.0.99.254

```

18.2 Administration SSH (restreinte)

Procedure

```

1 # Depuis VLAN ADMIN (doit ouvrir une session)
2 ssh admin@10.0.99.1
3
4 # Depuis un autre VLAN (doit echouer)
5 ssh admin@10.0.99.1

```


Preuves disponibles

Configuration (ACL + VTY) :

```

1 ip access-list standard SSH_ADMIN
2   permit 10.10.10.0 0.0.0.255
3   deny   any
4   !
5 line vty 0 4
6   access-class SSH_ADMIN in
7   login local
8   transport input ssh
9 line vty 5 15
10  access-class SSH_ADMIN in
11  login local
12  transport input ssh
13  !
14 line vty 5 15
15  access-class SSH_ADMIN in
16  login local
17  transport input ssh

```

Runtime (ACL) :

```

1 HQ-SW1#show access-lists SSH_ADMIN
2 Standard IP access list SSH_ADMIN
3   10 permit 10.0.10.0, wildcard bits 0.0.0.255
4   20 permit 10.0.99.0, wildcard bits 0.0.0.255
5   30 deny   any (3 matches)
6   40 permit 10.10.10.0, wildcard bits 0.0.0.255
7 HQ-SW1#

```

Journal (tentative bloquee) :

```

1 *Feb  4 23:14:20.924: %SEC-6-IPACCESSLOGP: list LAN_TO_PROXY denied tcp 10.10.10.50(59932) ->
  10.0.99.1(22), 1 packet

```

A produire en seance (validation complete)

```

1 show run | section line vty
2 show access-lists SSH_ADMIN
3 show logging | include SSH_ADMIN

```

18.3 Segmentation (PERSONNEL / PROD / WIFI / VIDEO)

Preuves disponibles (structure)

VLAN declares :

```

1 Q-SW2#show vlan brief
2
3 VLAN Name                Status    Ports
4 -----
5 1    default                active    Gi1/2, Gi1/3, Gi2/0, Gi2/1
6                                     Gi2/2, Gi2/3, Gi3/0, Gi3/1
7                                     Gi3/2, Gi3/3
8 10   VLAN10_ADMIN           active
9 20   VLAN20_PERSONNEL       active
10 30   VLAN30_VIDEO           active
11 40   VLAN40_PROD            active
12 50   VLAN50_WIFI_PERSO      active
13 60   VLAN60_WIFI_GUEST      active
14 70   VLAN70_SERVEURS        active

```

```

15 80 VLAN80_DMZ active
16 99 VLAN99_MGMT active Gi0/0, Gi1/0
17 1002 fddi-default act/unsup
18 1003 token-ring-default act/unsup
19 1004 fddinet-default act/unsup
20 1005 trnet-default

```

SVI up/up :

```

1 1 10.10.10.2 YES manual up up
2 2 10.10.20.2 YES manual up up
3 3 10.10.30.2 YES manual up up
4 4 10.10.40.2 YES manual up up
5 5 10.10.50.2 YES manual up up
6 6 10.10.60.2 YES manual up up
7 7 10.10.70.2 YES manual up up
8 8 10.10.80.2 YES manual up up
9 9 10.0.99.2 YES NVRAM up up
10
11
12 10.10.70.1 YES manual up up
13 10.10.80.1 YES manual up up
14 10.0.99.1 YES NVRAM up up
15
16
17 10.0.99.11 YES manual up up

```

A produire en seance (fonctionnel)

```

1 # Sur un hote de chaque VLAN evalue
2 ip a
3 ip route
4 ping -c 3 <VIP_VLAN>
5 ping -c 3 <IP_SERVICE>

```

18.4 Proxy HTTP

Preuves disponibles

Autorisation reseau (ACL runtime) :

```

1 HQ-SW2#show access-lists LAN_TO_PROXY
2 Extended IP access list LAN_TO_PROXY
3 10 permit udp any 224.0.0.0 0.0.0.255 (46 matches)
4 20 permit icmp any any
5 30 permit udp host 10.10.10.1 host 10.10.70.10 eq bootps
6 40 permit udp host 10.10.10.2 host 10.10.70.10 eq bootps
7 50 permit udp host 10.10.70.10 host 10.10.10.1 eq bootps
8 60 permit udp host 10.10.70.10 host 10.10.10.2 eq bootps
9 70 permit udp any eq bootps any eq bootpc
10 80 permit tcp 10.10.0.0 0.0.255.255 host 10.10.80.10 eq 3128
11 90 deny ip any any log (1 match)

```

A produire en seance (preuve applicative)

```

1 # Sur le serveur proxy
2 systemctl status squid --no-pager
3 ss -tlnp | grep 3128
4 tail -n 20 /var/log/squid/access.log
5

```

```

6 # Sur un client
7 export http_proxy="http://10.10.80.10:3128"
8 curl -I http://deb.debian.org

```

18.5 DNS

Preuves disponibles

Service BIND actif :

```

1 named.service - BIND Domain Name Server
2   Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enable>
3   Active: active (running) since Wed 2026-02-04 17:35:44 UTC; 32s ago
4     Docs: man:named(8)
5   Main PID: 415 (named)
6     Status: "running"
7     Tasks: 5 (limit: 537)
8    Memory: 25.5M
9       CPU: 1.417s
10    CGroup: /system.slice/named.service
11           415 /usr/

```

Capture de test (etat actuel) :

```

1 dig @127.0.0.1 dns.hq.local
2 dig @127.0.0.1 google.com
3
4 ; <<>> DiG 9.18.44-1-debi2u1-Debian <<>> @127.0.0.1 dns.hq.local
5 ; (1 server found)
6 ;; global options: +cmd
7 ;; Got answer:
8 ;; WARNING: .local is reserved for Multicast DNS
9 ;; You are currently testing what happens when an mDNS query is leaked to DNS
10 ;; ->>HEADER<<- opcode: QUERY, status: REFUSED, id: 34831
11 ;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
12 ;; WARNING: recursion requested but not available
13
14 ;; OPT PSEUDOSECTION:

```

A produire en seance (preuve conforme)

```

1 dig @<IP_DNS> dns.hq.local A
2 dig @<IP_DNS> google.com A
3 # Attendu : status: NOERROR + ANSWER

```

18.6 DHCP

Preuve disponible (client)

```

1 debian@debian:~$ sudo dhclient -v ens4
2 ip a
3 ip route
4 ping gw.hq.local
5 ping google.com
6 Internet Systems Consortium DHCP Client 4.4.3-P1
7 Copyright 2004-2022 Internet Systems Consortium.
8 All rights reserved.
9 For info, please visit https://www.isc.org/software/dhcp/
10

```

```

11 Listening on LPF/ens4/0c:ad:6c:13:00:00
12 Sending on LPF/ens4/0c:ad:6c:13:00:00
13 Sending on Socket/fallback
14 DHCPDISCOVER on ens4 to 255.255.255.255 port 67 interval 5
15 DHCPPOFFER of 10.0.10.50 from 10.0.10.10
16 DHCPREQUEST for 10.0.10.50 on ens4 to 255.255.255.255 port 67
17 DHCPACK of 10.0.10.50 from 10.0.10.10
18 bound to 10.0.10.50 -- renewal in 245 seconds.

```

A produire en seance (complement)

```

1 # Sur le client
2 sudo dhclient -r ens4
3 sudo dhclient -v ens4
4 ip a
5 ip route
6 cat /etc/resolv.conf
7
8 # Sur le serveur DHCP
9 sudo journalctl -u isc-dhcp-server -n 50 --no-pager

```

18.7 Redondance au siege (HSRP + bascule)

Preuve disponible (etat HSRP)

```

1 HQ-SW2#show standby brief
2                               P indicates configured to preempt.
3                               |
4 Interface    Grp  Pri P State   Active        Standby        Virtual IP
5 V110         10   100 P Standby  10.10.10.1    local          10.10.10.254
6 V120         20   100 P Standby  10.10.20.1    local          10.10.20.254
7 V130         30   100 P Standby  10.10.30.1    local          10.10.30.254
8 V140         40   100 P Standby  10.10.40.1    local          10.10.40.254
9 V150         50   100 P Standby  10.10.50.1    local          10.10.50.254
10 V160         60   100 P Standby  10.10.60.1    local          10.10.60.254
11 V170         70   100 P Standby  10.10.70.1    local          10.10.70.254
12 V180         80   100 P Standby  10.10.80.1    local          10.10.80.254
13 V199         99   100 P Standby  10.0.99.1     local          10.0.99.254

```

A produire en seance (bascule)

```

1 # Couper un lien / interface critique (ou simuler une panne)
2 # puis verifier le changement d'etat :
3 show standby brief
4
5 # Sur un hote :
6 ping -c 5 10.10.10.254

```

18.8 VPN siege-succursale

Preuve disponible (captures)

```

1 Router2#show crypto isakmp sa
2 IPv4 Crypto ISAKMP SA
3 dst          src          state          conn-id status
4

```

```
5 IPv6 Crypto ISAKMP SA
6
7 Router2#show crypto ipsec sa
8
9 interface: GigabitEthernet0/3
10   Crypto map tag: CMAP-SITE, local addr 10.10.250.10
11
12   protected vrf: VRF-SITE
13   local ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
14   remote ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
15   current_peer 10.10.250.18 port 500
16     PERMIT, flags={origin_is_acl,}
17     #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
18     #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
19     #pkts compressed: 0, #pkts decompressed: 0
20     #pkts not compressed: 0, #pkts compr. failed: 0
21     #pkts not decompressed: 0, #pkts decompress failed: 0
22     #send errors 0, #recv errors 0
23
24   local crypto endpt.: 10.10.250.10, remote crypto endpt.: 10.10.250.18
25
26 Router2#ping vrf VRF-SITE 10.20.0.1
27 Type escape sequence to abort.
28 Sending 5, 100-byte ICMP Echos to 10.20.0.1, timeout is 2 seconds:
29 .....
30 Success rate is 0 percent (0/5)
31 Router2#
```

A produire en seance (tunnel + compteurs)

```
1 show crypto isakmp sa
2 show crypto ipsec sa
3 # Generer du trafic inter-sites :
4 ping -c 5 <IP_DISTANT>
5 # Reverifier :
6 show crypto ipsec sa
```

18.9 Centralisation des logs

Preuves disponibles

Configuration (equipements) :

```
1 logging source-interface Vlan99
2 logging host 10.0.10.10
3 service timestamps debug datetime msec
4 service timestamps log datetime msec
```

Service rsyslog actif :

```
1 * rsyslog.service - System Logging Service
2   Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
3   Active: active (running) since Wed 2026-02-04 17:44:17 UTC; 504ms ago
4   TriggeredBy: * syslog.socket
5     Docs: man:rsyslogd(8)
6           man:rsyslog.conf(5)
7           https://www.rsyslog.com/doc/
8   Main PID: 528 (rsyslogd)
9     Tasks: 7 (limit: 537)
10    Memory: 908.0K
11      CPU: 133ms
12    CGroup: /system.slice/rsyslog.service
13           528 /usr/sbin/rsyslogd -n -iNONE
```

A produire en seance (preuve de reception)

```
1 # Cote equipement (generer un evenement)
2 conf t
3 interface Gi0/2
4   shutdown
5   no shutdown
6 end
7
8 # Cote serveur Syslog
9 sudo tail -f /var/log/syslog
10 # ou
11 sudo journalctl -u rsyslog -f
```

19 Conclusion

La maquette couvre la segmentation logique multi-sites, la redondance de passerelle au siege et le socle de services (DHCP/DNS/proxy/logs). Les annexes rassemblent les configurations et les extraits de preuves disponibles. Les procedures de tests permettent de produire, lors de la seance d'évaluation, les demonstrations attendues pour les points dont les captures n'étaient pas presentes dans les elements fournis.

20 Annexes

21 Annexes

22 Annexe A - Extraits de configuration (cibles)

Lisibilité

Les configurations complètes peuvent être volumineuses (bannières, sections non pertinentes). Cette annexe regroupe uniquement les extraits nécessaires à l'évaluation. Les fichiers complets restent disponibles dans `annexes/configs/`.

22.1 SVI / VLAN (siege)

```
1 interface Vlan10
2   ip address 10.10.10.1 255.255.255.0
3   ip helper-address 10.10.70.10
4   standby 10 ip 10.10.10.254
5   standby 10 priority 110
6   standby 10 preempt
7   !
8 interface Vlan20
9   ip address 10.10.20.1 255.255.255.0
10  ip helper-address 10.10.70.10
11  standby 20 ip 10.10.20.254
12  standby 20 priority 110
13  standby 20 preempt
14  !
15 interface Vlan30
16  ip address 10.10.30.1 255.255.255.0
17  ip helper-address 10.10.70.10
18  standby 30 ip 10.10.30.254
19  standby 30 priority 110
20  standby 30 preempt
21  !
22 interface Vlan40
23  ip address 10.10.40.1 255.255.255.0
24  ip helper-address 10.10.70.10
25  standby 40 ip 10.10.40.254
26  standby 40 priority 110
27  standby 40 preempt
28  !
29 interface Vlan50
30  ip address 10.10.50.1 255.255.255.0
31  ip helper-address 10.10.70.10
32  standby 50 ip 10.10.50.254
33  standby 50 priority 110
34  standby 50 preempt
35  !
36 interface Vlan60
37  ip address 10.10.60.1 255.255.255.0
38  ip helper-address 10.10.70.10
39  standby 60 ip 10.10.60.254
40  standby 60 priority 110
41  standby 60 preempt
42  !
43 interface Vlan70
44  ip address 10.10.70.1 255.255.255.0
45  ip helper-address 10.10.70.10
46  standby 70 ip 10.10.70.254
47  standby 70 priority 110
48  standby 70 preempt
49  !
50 interface Vlan80
51  ip address 10.10.80.1 255.255.255.0
```

```
52 ip access-group DMZ_IN in
53 ip helper-address 10.10.70.10
54 standby 80 ip 10.10.80.254
55 standby 80 priority 110
56 standby 80 preempt
57 !
58 interface Vlan99
59 ip address 10.0.99.1 255.255.255.0
60 standby 99 ip 10.0.99.254
61 standby 99 priority 110
62 standby 99 preempt
```

22.2 HSRP (siege)

```
1 interface Vlan10
2 ip address 10.10.10.1 255.255.255.0
3 ip helper-address 10.10.70.10
4 standby 10 ip 10.10.10.254
5 standby 10 priority 110
6 standby 10 preempt
7 !
8 interface Vlan20
9 ip address 10.10.20.1 255.255.255.0
10 ip helper-address 10.10.70.10
11 standby 20 ip 10.10.20.254
12 standby 20 priority 110
13 standby 20 preempt
14 !
15 interface Vlan30
16 ip address 10.10.30.1 255.255.255.0
17 ip helper-address 10.10.70.10
18 standby 30 ip 10.10.30.254
19 standby 30 priority 110
20 standby 30 preempt
21 !
22 interface Vlan40
23 ip address 10.10.40.1 255.255.255.0
24 ip helper-address 10.10.70.10
25 standby 40 ip 10.10.40.254
26 standby 40 priority 110
27 standby 40 preempt
28 !
29 interface Vlan50
30 ip address 10.10.50.1 255.255.255.0
31 ip helper-address 10.10.70.10
32 standby 50 ip 10.10.50.254
33 standby 50 priority 110
34 standby 50 preempt
35 !
36 interface Vlan60
37 ip address 10.10.60.1 255.255.255.0
38 ip helper-address 10.10.70.10
39 standby 60 ip 10.10.60.254
40 standby 60 priority 110
41 standby 60 preempt
42 !
43 interface Vlan70
44 ip address 10.10.70.1 255.255.255.0
45 ip helper-address 10.10.70.10
46 standby 70 ip 10.10.70.254
47 standby 70 priority 110
48 standby 70 preempt
49 !
50 interface Vlan80
51 ip address 10.10.80.1 255.255.255.0
52 ip access-group DMZ_IN in
53 ip helper-address 10.10.70.10
54 standby 80 ip 10.10.80.254
55 standby 80 priority 110
56 standby 80 preempt
57 !
```



```
58 interface Vlan99
59   ip address 10.0.99.1 255.255.255.0
60   standby 99 ip 10.0.99.254
61   standby 99 priority 110
62   standby 99 preempt
63   !
64 interface Vlan10
65   ip address 10.10.10.2 255.255.255.0
66   ip access-group LAN_TO_PROXY in
67   ip helper-address 10.10.70.10
68   standby 10 ip 10.10.10.254
69   standby 10 preempt
70   !
71 interface Vlan20
72   ip address 10.10.20.2 255.255.255.0
73   ip access-group LAN_TO_PROXY in
74   ip helper-address 10.10.70.10
75   standby 20 ip 10.10.20.254
76   standby 20 preempt
77   !
78 interface Vlan30
79   ip address 10.10.30.2 255.255.255.0
80   ip access-group LAN_TO_PROXY in
81   ip helper-address 10.10.70.10
82   standby 30 ip 10.10.30.254
83   standby 30 preempt
84   !
85 interface Vlan40
86   ip address 10.10.40.2 255.255.255.0
87   ip access-group LAN_TO_PROXY in
88   ip helper-address 10.10.70.10
89   standby 40 ip 10.10.40.254
90   standby 40 preempt
91   !
92 interface Vlan50
93   ip address 10.10.50.2 255.255.255.0
94   ip access-group LAN_TO_PROXY in
95   ip helper-address 10.10.70.10
96   standby 50 ip 10.10.50.254
97   standby 50 preempt
98   !
99 interface Vlan60
100   ip address 10.10.60.2 255.255.255.0
101   ip access-group LAN_TO_PROXY in
102   ip helper-address 10.10.70.10
103   standby 60 ip 10.10.60.254
104   standby 60 preempt
105   !
106 interface Vlan70
107   ip address 10.10.70.2 255.255.255.0
108   ip helper-address 10.10.70.10
109   standby 70 ip 10.10.70.254
110   standby 70 preempt
111   !
112 interface Vlan80
113   ip address 10.10.80.2 255.255.255.0
114   ip access-group DMZ_IN in
115   ip helper-address 10.10.70.10
116   standby 80 ip 10.10.80.254
117   standby 80 preempt
118   !
119 interface Vlan99
120   ip address 10.0.99.2 255.255.255.0
121   standby 99 ip 10.0.99.254
122   standby 99 preempt
```

22.3 STP (siege)

```
1 spanning-tree mode rapid-pvst
2 spanning-tree extend system-id
3 spanning-tree vlan 10,20,30,40,50,60,70,80,99 priority 24576
```

22.4 SSH (ACL + VTY)

```
1 ip access-list standard SSH_ADMIN
2   permit 10.10.10.0 0.0.0.255
3   deny   any
4   !
5 line vty 0 4
6   access-class SSH_ADMIN in
7   login local
8   transport input ssh
9 line vty 5 15
10  access-class SSH_ADMIN in
11  login local
12  transport input ssh
13  !
14 line vty 5 15
15  access-class SSH_ADMIN in
16  login local
17  transport input ssh
```

22.5 DMZ / Proxy (ACL)

```
1 ip access-list extended DMZ_IN
2   permit tcp any any established
3   permit tcp host 10.10.80.10 any eq www
4   permit tcp host 10.10.80.10 any eq 443
5   permit udp host 10.10.80.10 any eq domain
6   permit tcp 10.10.10.0 0.0.0.255 host 10.10.80.10 eq 3128
7   permit tcp 10.10.30.0 0.0.0.255 host 10.10.80.10 eq 3128
8   permit tcp 10.10.20.0 0.0.0.255 host 10.10.80.10 eq 3128
9   permit udp host 10.10.80.10 host 10.10.70.10 eq domain
10  deny   ip any any log
11  remark === DNS vers serveur interne ===
12  permit udp 10.10.80.0 0.0.0.255 host 10.10.70.10 eq domain
13  permit tcp 10.10.80.0 0.0.0.255 host 10.10.70.10 eq domain
14  remark === HTTP/HTTPS vers Internet ===
15  permit tcp 10.10.80.0 0.0.0.255 any eq www
16  permit tcp 10.10.80.0 0.0.0.255 any eq 443
17  remark === ICMP debug (optionnel) ===
18  permit udp host 10.10.80.10 any eq ntp
19 ip access-list extended LAN_TO_PROXY
20  remark === Multicast (HSRP, routing interne)
21  permit udp any 224.0.0.0 0.0.0.255
22  remark === ICMP (debug, supervision)
23  permit icmp any any
24  remark === DHCP (clients)
25  permit udp any eq bootpc any eq bootps
26  permit udp any eq bootps any eq bootpc
27  remark === DHCP relay vers serveur
28  permit udp host 10.10.10.1 host 10.10.70.10 eq bootps
29  permit udp host 10.10.10.2 host 10.10.70.10 eq bootps
30  permit udp host 10.10.70.10 host 10.10.10.1 eq bootps
31  permit udp host 10.10.70.10 host 10.10.10.2 eq bootps
32  remark === DNS vers serveur interne
33  permit udp 10.10.0.0 0.0.255.255 host 10.10.70.10 eq domain
34  permit tcp 10.10.0.0 0.0.255.255 host 10.10.70.10 eq domain
35  remark === SSH ADMIN (ALLER)
36  permit tcp 10.10.10.0 0.0.0.255 10.0.99.0 0.0.0.255 eq 22
37  remark === SSH ADMIN (RETOUR)
38  permit tcp 10.0.99.0 0.0.0.255 10.10.10.0 0.0.0.255 established
39  remark === Proxy Squid
40  permit tcp 10.10.0.0 0.0.255.255 host 10.10.80.10 eq 3128
41  remark === DENY FINAL
42  deny   ip any any log
```

22.6 Syslog

```
1 logging source-interface Vlan99
2 logging host 10.0.10.10
3 service timestamps debug datetime msec
4 service timestamps log datetime msec
```

22.7 VPN IPsec

```
1 crypto isakmp policy 10
2   encr aes
3   authentication pre-share
4   group 14
5 crypto isakmp key VPN-SITE address 10.10.250.18
6 !
7 crypto isakmp key VPN-SITE address 10.10.250.18
8 !
9 crypto ipsec transform-set TS-SITE esp-aes esp-sha-hmac
10  mode tunnel
11 !
12 crypto map CMAP-SITE 10 ipsec-isakmp
13   set peer 10.10.250.18
14   set transform-set TS-SITE
15   match address VPN-SITE
16 !
17 ip access-list extended VPN-SITE
18   permit ip 10.10.0.0 0.0.255.255 10.20.0.0 0.0.255.255
```

22.8 Succursale (sous-interfaces)

```
1 interface GigabitEthernet0/1.10
2   encapsulation dot1Q 10
3   ip address 10.20.10.1 255.255.255.0
4 !
5 interface GigabitEthernet0/1.20
6   encapsulation dot1Q 20
7   ip address 10.20.20.1 255.255.255.0
8 !
9 interface GigabitEthernet0/1.30
10  encapsulation dot1Q 30
11  ip address 10.20.30.1 255.255.255.0
12 !
13 interface GigabitEthernet0/1.40
14  encapsulation dot1Q 40
15  ip address 10.20.40.1 255.255.255.0
16 !
17 interface GigabitEthernet0/1.50
18  encapsulation dot1Q 50
19  ip address 10.20.50.1 255.255.255.0
20 !
21 interface GigabitEthernet0/1.60
22  encapsulation dot1Q 60
23  ip address 10.20.60.1 255.255.255.0
24 !
25 interface GigabitEthernet0/1.99
26  encapsulation dot1Q 99
27  ip address 10.20.99.1 255.255.255.0
```

22.9 Routage (routes)

```
1 ip route 0.0.0.0 0.0.0.0 10.10.250.18
2 ip route 0.0.0.0 0.0.0.0 10.10.250.9
3 ip route vrf VRF-SITE 0.0.0.0 0.0.0.0 10.10.250.9
4 ip route 0.0.0.0 0.0.0.0 10.10.250.17
5 ip route 10.10.0.0 255.255.0.0 GigabitEthernet0/0
```

23 Annexe B - Extraits de preuves

Note

Ces extraits proviennent des sorties de commandes presentes dans les elements fournis.
Aucun resultat n'est invente.

23.1 VLAN

```

1 Q-SW2#show vlan brief
2
3 VLAN Name                               Status   Ports
4 -----
5 1    default                             active   Gi1/2, Gi1/3, Gi2/0, Gi2/1
6                                           Gi2/2, Gi2/3, Gi3/0, Gi3/1
7                                           Gi3/2, Gi3/3
8 10   VLAN10_ADMIN                        active
9 20   VLAN20_PERSONNEL                   active
10 30   VLAN30_VIDEO                       active
11 40   VLAN40_PROD                        active
12 50   VLAN50_WIFI_PERSO                  active
13 60   VLAN60_WIFI_GUEST                   active
14 70   VLAN70_SERVEURS                     active
15 80   VLAN80_DMZ                         active
16 99   VLAN99_MGMT                        active   Gi0/0, Gi1/0
17 1002 fddi-default                       act/unsup
18 1003 token-ring-default                  act/unsup
19 1004 fddinet-default                     act/unsup
20 1005 trnet-default

```

23.2 Interfaces VLAN (SVI)

```

1 Vlan10      10.10.10.2    YES manual up      up
2 Vlan20      10.10.20.2    YES manual up      up
3 Vlan30      10.10.30.2    YES manual up      up
4 Vlan40      10.10.40.2    YES manual up      up
5 Vlan50      10.10.50.2    YES manual up      up
6 Vlan60      10.10.60.2    YES manual up      up
7 Vlan70      10.10.70.2    YES manual up      up
8 Vlan80      10.10.80.2    YES manual up      up
9 Vlan99      10.0.99.2      YES NVRAM up        up
10
11
12 Vlan70      10.10.70.1    YES manual up      up
13 Vlan80      10.10.80.1    YES manual up      up
14 Vlan99      10.0.99.1      YES NVRAM up        up
15
16
17 Vlan99      10.0.99.11     YES manual up      up

```

23.3 HSRP

```

1 HQ-SW2#show standby brief
2           P indicates configured to preempt.
3           |
4 Interface  Grp  Pri P State   Active          Standby          Virtual IP
5 V110      10   100 P Standby  10.10.10.1      local            10.10.10.254
6 V120      20   100 P Standby  10.10.20.1      local            10.10.20.254
7 V130      30   100 P Standby  10.10.30.1      local            10.10.30.254
8 V140      40   100 P Standby  10.10.40.1      local            10.10.40.254
9 V150      50   100 P Standby  10.10.50.1      local            10.10.50.254
10 V160      60   100 P Standby  10.10.60.1      local            10.10.60.254
11 V170      70   100 P Standby  10.10.70.1      local            10.10.70.254

```

12	V180	80	100 P Standby	10.10.80.1	local	10.10.80.254
13	V199	99	100 P Standby	10.0.99.1	local	10.0.99.254

23.4 SSH (ACL + journal)

```
1 HQ-SW1#show access-lists SSH_ADMIN
2 Standard IP access list SSH_ADMIN
3     10 permit 10.0.10.0, wildcard bits 0.0.0.255
4     20 permit 10.0.99.0, wildcard bits 0.0.0.255
5     30 deny any (3 matches)
6     40 permit 10.10.10.0, wildcard bits 0.0.0.255
7 HQ-SW1#
```

```
1 *Feb  4 23:14:20.924: %SEC-6-IPACCESSLOGP: list LAN_TO_PROXY denied tcp 10.10.10.50(59932) ->
    10.0.99.1(22), 1 packet
```

23.5 Proxy : ACL d'accès

```
1 HQ-SW2#show access-lists LAN_TO_PROXY
2 Extended IP access list LAN_TO_PROXY
3     10 permit udp any 224.0.0.0 0.0.0.255 (46 matches)
4     20 permit icmp any any
5     30 permit udp host 10.10.10.1 host 10.10.70.10 eq bootps
6     40 permit udp host 10.10.10.2 host 10.10.70.10 eq bootps
7     50 permit udp host 10.10.70.10 host 10.10.10.1 eq bootps
8     60 permit udp host 10.10.70.10 host 10.10.10.2 eq bootps
9     70 permit udp any eq bootps any eq bootpc
10    80 permit tcp 10.10.0.0 0.0.255.255 host 10.10.80.10 eq 3128
11    90 deny ip any any log (1 match)
```

23.6 DMZ : logs d'ACL

```
1 *Feb  4 22:37:12.228: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(37866) ->
    176.137.36.37(123), 1 packet
2 *Feb  4 22:37:22.500: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(38739) -> 5.39.80.51(123),
    1 packet
3 *Feb  4 22:37:32.703: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(58751) ->
    141.95.158.40(123), 1 packet
4 *Feb  4 22:37:39.458: %SEC-6-IPACCESSLOGR: access-list logging rate-limited or missed 4 packets
5 *Feb  4 22:37:42.431: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.1(1985) -> 224.0.0.2(1985),
    134 packets
6 *Feb  4 23:07:18.531: %SEC-6-IPACCESSLOGP: list DMZ_IN denied udp 10.10.80.10(44903) -> 82.66.40.79(123),
    1 packet
```

23.7 DHCP

```
1 debian@debian:~$ sudo dhclient -v ens4
2 ip a
3 ip route
4 ping gw.hq.local
5 ping google.com
6 Internet Systems Consortium DHCP Client 4.4.3-P1
7 Copyright 2004-2022 Internet Systems Consortium.
8 All rights reserved.
9 For info, please visit https://www.isc.org/software/dhcp/
10
11 Listening on LPF/ens4/0c:ad:6c:13:00:00
12 Sending on LPF/ens4/0c:ad:6c:13:00:00
13 Sending on Socket/fallback
14 DHCPDISCOVER on ens4 to 255.255.255.255 port 67 interval 5
15 DHCPOFFER of 10.0.10.50 from 10.0.10.10
```

```
16 DHCPREQUEST for 10.0.10.50 on ens4 to 255.255.255.255 port 67
17 DHCPACK of 10.0.10.50 from 10.0.10.10
18 bound to 10.0.10.50 -- renewal in 245 seconds.
```

23.8 DNS (etat)

```
1 named.service - BIND Domain Name Server
2   Loaded: loaded (/lib/systemd/system/named.service; enabled; preset: enable>
3   Active: active (running) since Wed 2026-02-04 17:35:44 UTC; 32s ago
4     Docs: man:named(8)
5   Main PID: 415 (named)
6     Status: "running"
7     Tasks: 5 (limit: 537)
8   Memory: 25.5M
9     CPU: 1.417s
10    CGroup: /system.slice/named.service
11           415 /usr/
```

```
1 dig @127.0.0.1 dns.hq.local
2 dig @127.0.0.1 google.com
3
4 ; <<>> DiG 9.18.44-1~deb12u1-Debian <<>> @127.0.0.1 dns.hq.local
5 ; (1 server found)
6 ;; global options: +cmd
7 ;; Got answer:
8 ;; WARNING: .local is reserved for Multicast DNS
9 ;; You are currently testing what happens when an mDNS query is leaked to DNS
10 ;; ->>HEADER<- opcode: QUERY, status: REFUSED, id: 34831
11 ;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1
12 ;; WARNING: recursion requested but not available
13
14 ;; OPT PSEUDOSECTION:
```

23.9 Syslog (service rsyslog)

```
1 * rsyslog.service - System Logging Service
2   Loaded: loaded (/lib/systemd/system/rsyslog.service; enabled; preset: enabled)
3   Active: active (running) since Wed 2026-02-04 17:44:17 UTC; 504ms ago
4 TriggeredBy: * syslog.socket
5     Docs: man:rsyslogd(8)
6           man:rsyslog.conf(5)
7           https://www.rsyslog.com/doc/
8   Main PID: 528 (rsyslogd)
9     Tasks: 7 (limit: 537)
10  Memory: 908.0K
11    CPU: 133ms
12    CGroup: /system.slice/rsyslog.service
13           528 /usr/sbin/rsyslogd -n -iNONE
```

23.10 VPN (captures)

```
1 Router2#show crypto isakmp sa
2 IPv4 Crypto ISAKMP SA
3 dst          src          state          conn-id status
4
5 IPv6 Crypto ISAKMP SA
6
7 Router2#show crypto ipsec sa
8
9 interface: GigabitEthernet0/3
10   Crypto map tag: CMAP-SITE, local addr 10.10.250.10
11
12   protected vrf: VRF-SITE
13   local ident (addr/mask/prot/port): (10.10.0.0/255.255.0.0/0/0)
```

```
14 remote ident (addr/mask/prot/port): (10.20.0.0/255.255.0.0/0/0)
15 current_peer 10.10.250.18 port 500
16 PERMIT, flags={origin_is_acl,}
17 #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
18 #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
19 #pkts compressed: 0, #pkts decompressed: 0
20 #pkts not compressed: 0, #pkts compr. failed: 0
21 #pkts not decompressed: 0, #pkts decompress failed: 0
22 #send errors 0, #recv errors 0
23
24 local crypto endpt.: 10.10.250.10, remote crypto endpt.: 10.10.250.18
25
26 Router2#ping vrf VRF-SITE 10.20.0.1
27 Type escape sequence to abort.
28 Sending 5, 100-byte ICMP Echos to 10.20.0.1, timeout is 2 seconds:
29 .....
30 Success rate is 0 percent (0/5)
31 Router2#
```